Review 5.0

Tuesday, 8 April 2025 9:12 AM

Conservative & Non-conservative

Equations obtained from directly applying fundamental physical principles to a finite control volume

↓

i.e. eq. in integral form (that are then applied to obtain PDE's)

Entire system of governing equations in conservation form } continuity, momentum & energy

unsteady (solution vector) | flux terms | source term

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} + \frac{\partial H}{\partial z} = S$$

"Shock" → sharp, discontinuous changes in the primitive flow-field variables (p, ρ, u, T)

↳ Conservation form of equations should be used to get smooth + stable (or else you get wiggles

↳ because conservation form in fact uses (ρu) & product of dependent variables ↳ which are generally conserved in a shock!

(i.e. changes in flux are either zero or small across a shock wave)

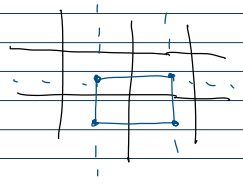
Office hours:

+ Two-boundary method for grid generation

↓

Jacobian → Area in 2D

↳ Volume in 3D



↳ Local Thermodynamic equilibrium

↓

When can you assume it?

If it is changing

→ Equations of state ($PV = nRT$)

→ connect measurable/non-measurable variables

↓ ↳ energy (

T, ρ

↳ Gaussian elimination (Accuracy)

↳ Condition no.

↳ residual

↳ Iterative refinement →

↳ Modified equation & phase angles

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ϕ - phase angle relates to odd derivatives terms

↳ Implicit vs. Explicit ✓

↳ FDE's

↳ Errors

Types of PDE's & their significance:

$u = u(x,y)$

$v = v(x,y)$

$\frac{\partial u}{\partial x}$ $\frac{\partial u}{\partial y}$

$\frac{\partial v}{\partial x}$ $\frac{\partial v}{\partial x}$

Making matrix & checking determinant/ eigenvalues

$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$

$dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy$

characteristic lines

↓

derivative of u & v are indeterminant

(i.e. derivative of solution do not have a unique solution)

↓

PDE's simplify to ODE's

→ Gives type of PDE's

For first order : hyperbolic, parabolic, elliptic (quasilinear)

information at a given point P influences only those regions b/w advancing characteristics

information at point P in the xy-plane influences the entire region of the plane to one side of P

info at P influences all other regions of the domain

↓

entire boundary must be defined.

eg:

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2 + p)}{\partial x} = 0$$